

Hacks and tweaks

Are your rainbows mutant enough? No? Here are some ideas you could try to make them even more bizarre!

▽ Colour surprise!

In Python, colours can also be described by using RGB values – this stands for red, green, blue. Choosing values at random for the amounts of red, green, and blue in a colour means the colour itself will be completely random. Try replacing the code in the `move_turtle()` function with some new code that uses RGB values instead of colour names. Now run the code to see what colours appear!

```
pen_colors = ['red', 'orange', 'yellow', 'green', 'blue', 'purple']
t.pencolor(random.choice(pen_colors))
```

Replace these two lines with...

```
t.colormode(255)
red = random.randint(0, 255)
blue = random.randint(0, 255)
green = random.randint(0, 255)
t.pencolor(red, green, blue)
```

...these five lines.

▽ Mix up the lines

Don't just stick to one width for the line – draw even more scrambled rainbows with this hack! The lines will change at random from really thin to superthick and all widths in between. Add this code to the `move_turtle()` function after you set `t.pencolor`.

```
t.pensize(random.randint(1,40))
```



EXPERT TIPS

RGB colours

In **turtle**, the colour “blue” is (0, 0, 255) in RGB values, as it’s made up of the maximum amount of blue, with no red or green. If you want to use RGB values for the turtle’s pen colour, you need to let Python know by using the command `t.colormode(255)`, or it will expect a string and give you an error.

This number shows the amount of red in the colour (between 0 and 255).

```
t.pencolor(0, 0, 255)
```

The amount of green

The amount of blue

```
t.pencolor('blue')
```